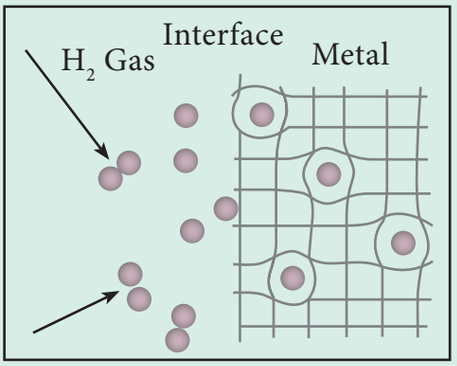
METAL HYDRIDE (HYDROGEN SPONGE)





The best studied binary hydrides are the palladium-hydrogen system. Hydrogen interacts with palladium in a unique way, and forms a limiting monohydride, PdH. Upon heating, H atoms diffuse through the metal to the surface and recombine to form molecular hydrogen. Since no other gas behaves this way with palladium, this process has been used to separate hydrogen gas from other gases:

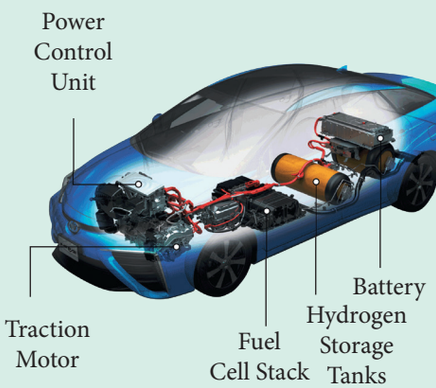
$$2\ Pd\ (s) + H_2\ (g) \leftarrow 2\ PdH\ (s).$$

The hydrogen molecule readily adsorb on the palladium surface, where it dissociates into atomic hydrogen. The dissociated atoms dissolve into the interstices or voids (octahedral/tetrahedral) of the crystal lattice.

Technically, the formation of metal hydride is by chemical reaction but it behaves like a physical storage method, i.e., it is absorbed and released like a water sponge. Such a reversible uptake of hydrogen in metals and alloys is also attractive for hydrogen storage and for rechargeable metal hydride battery applications.

The Hydrogen – Future Fuel





The depletion of fossil fuel reserves and the threat of global warming make the world to shift toward alternative fuels. In this regard, hydrogen is often considered as a potential candidate for this purpose as it is a clean burning fuel as it does not produce any pollutants upon burning. Hence, hydrogen can directly be used as a fuel and can replace existing gasoline (petrol)/diesel/kerosene powered engines, and/or indirectly be used with oxygen in fuel cells to generate electricity. However, there are some difficulties needs to be overcome. The production cost is high and also uses more fossil fuels. Due to its volatile and flammable nature in its native form, it is difficult to store and transport. Being the smallest molecule it is prone to leakage.

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